

Practical 2

Aim: Write a C++ program to implement bubble sort and display the sorted array.

Source-code:

(User-defined Array)

```
#include<iostream>
using namespace std;
int main()
{
    int n,arr[100];
    cout<<"Enter no of elements:";
    cin>>n;
    cout<<"Enter elements:\n";
    for(int i=0;i<n;i++){
        cin>>arr[i];
    }

    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-i-1;j++){
            if(arr[j]>arr[j+1]){
                int temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
    cout<<"Sorted array:";
    for(int i=0;i<n;i++){
        cout<<arr[i]<<" ";
    }
    return 0;
}
```

(Pre-defined Array)

```
#include<iostream>
using namespace std;
int main()
{
    int arr[]={15,16,6,8,5};
    int n=5;
    for(int i=0;i<n-1;i++){

        for(int j=0;j<i-1;j++){

            if(arr[j]>arr[j+1]){
                int temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
    cout<<"Sorted array:";
    for(int i=0;i<n;i++){
        cout<<arr[i]<<" ";
    }
    return 0;
}
```

Output:

(For User-defined Array)

```
Enter no of elements:4
```

```
Enter elements:
```

```
2 7 9 5
```

```
Sorted array:2 5 7 9
```

(For Pre-defined Array)

```
Original Array: 15 16 6 8 5
```

```
Sorted Array: 5 6 8 15 16
```